



An analysis of the reversible and irreversible degradation of a PEM fuel cell short-stack operated under simulated maritime conditions

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HIGHLIGHTS

- A commercial PEMFC short-stack was tested under simulated maritime conditions.
- Reversible degradation was found to dominate, particularly under constant operation.
- Irreversible degradation rates as low as $1.2 \mu\text{V h}^{-1}$ were recorded.
- Irreversible losses were due to a 20 % loss in cathode ECSA over the testing period.

ABSTRACT

A PEMFC short stack was operated for 1500 h under maritime-relevant conditions by applying alternating 125 h periods of constant load and a dynamic maritime load profile. The reversible and irreversible degradation rates were quantified, with the reversible degradation dominating for both segment types, but being around 2.5 times higher for the constant segments than the dynamic ($52 \mu\text{V h}^{-1}$ compared to 22). This implies that the reversible degradation should be actively mitigated regardless of type of vessel operation and hybridisation strategy, as it can lead to significant drops in the stack voltage and therefore efficiency. The average irreversible degradation increased from 1.2 to $4.8 \mu\text{V h}^{-1}$ when going from static to dynamic cycles. This degradation was found to be due to a cathode ECSA loss of around 20 %, with no notable increase in hydrogen crossover or HFR, indicating membrane stability during the testing period.

1. Introduction

The World Energy Outlook report from the International Energy Agency (IEA) in 2024 states that emissions worldwide are set to peak soon, but there is an increasingly narrow pathway to limiting global temperature rise to 1.5°C [1]. The shipping industry alone is responsible for emitting more than 700 Mt CO_2 /year (2022) [2], so replacing fossil fuels by clean green technologies in this sector is important. In 2023 the International Maritime Organization (IMO) published a strategy on the reduction of GHG emissions from ships, where they envisage a reduction in carbon intensity of international shipping by at least 40 % and total annual GHG emission reduction of at least 20 % by 2030 compared to 2008 numbers [3]. With only a 3.6 % reduction in GHG emissions by 2023, great effort is needed to reach the 2030 goals. One way of reaching these goals is through use of zero-emission technologies, fuels and/or energy sources, which the IEA states should constitute at least 10 % of the energy used by 2030.

Interest in using fuel cells for maritime applications is increasing, due to their relatively lightweight, low-volume drivetrain, high efficiency, and ability to adapt to variable power demands [4,5]. Fuel cells utilise hydrogen to produce power, with the only byproducts being water and heat. When utilising green hydrogen, the power produced from fuel cells is zero emission. Low-temperature Proton Exchange Membrane (PEM) fuel cells (FCs) have already been widely demonstrated for road and rail transport [4,6,7], and now several manufacturers are in the process of developing maritime fuel cell systems.

Compared to PEM FCs for other applications, maritime systems should have a longer lifetime, with goals for system lifetimes up to 80 000 h [8]. When such a long lifetime is needed, the total cost of ownership is dominated by the fuel cost rather than the initial system cost [9]. This means that the focus is on higher quality, more durable components, with a higher initial cost (CAPEX) but designed for a lower operational cost (OPEX), i.e. an improved efficiency. It has been shown that fuel cell systems have the highest efficiency at low power [4,10], so

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maritime fuel cell systems are likely to be operated at low to medium average power. In addition, energy management strategies for fuel cells for maritime systems can be complex, ranging between load levelling approaches where the fuel cell operates at a constant load, with a large hybrid system, to load following approaches where the fuel cell load directly follows the load demand and there is only a small or no battery hybridisation.

[11] However, historically, most of the literature and testing protocols available for fuel cell stack testing has been focused on automotive applications, which have highly dynamic operation and small battery systems [12–14]. This knowledge cannot be directly transferred to maritime fuel cells due to differences in load profiles, operation strategies, and hybridisation strategies, and there are currently no standardised maritime accelerated stress tests (ASTs) [15].

When FCs are operated for maritime applications, the relative contribution of reversible degradation compared to irreversible degradation increases. The main contributors to reversible degradation in PEM fuel cells are oxidation of the Pt catalyst, and buildup of water at the GDL/CL interface [13,16]. Strategies to mitigate this reversible degradation include applying a gas purge (water accumulation) or cycling the potential (Pt oxidation) through application of high currents, or by utilising techniques such as cyclic voltammetry [17].

There is currently little information available on the impact of reversible degradation on the long-term operation of fuel cells at constant current densities, especially at relatively low currents for long durations, as would be most relevant for maritime applications [18]. There have been some works that separate the reversible and irreversible contributions to degradation [13,19], but on the single cell level, or with a focus on automotive end-use [20,21], which differs from maritime use due to the highly dynamic load profiles. In one example, Chu et al. [17] investigated various strategies for mitigating the reversible degradation during long-term automotive-relevant stack testing but did not quantify the amount of reversible degradation observed. In other works, Ma et al. [20] and Li et al. [21] investigated the recoverable performance losses for a fuel cell stack under an automotive AST. In these cases, the authors disagree about whether the major contributor to reversible performance loss is platinum oxidation, or the buildup of water.

In this work, we will for the first time report quantification of both the reversible and irreversible performance losses incurred during operation of a commercial fuel cell short-stack under maritime load dynamics. The causes of irreversible performance degradation are investigated using electrochemical analysis combined with computational fitting, and an understanding of the differences in degradation when operating at constant currents, i.e. load levelling or high levels of hybridisation, compared to operating under a dynamic cycle i.e. load following or low levels of hybridisation is gained. This will help investigate the durability and dynamic response of PEMFCs whilst aiming for a minimum system OPEX, thus overcoming several key barriers to the implementation of fuel cells in maritime applications [9].

2. Methodology

A commercial 10-cell fuel cell short-stack (NedStack Fuel Cell Technology, Carbon Technology Energies B.V.) operated using a Greenlight G400 test station was used in this work. The gasses used were hydrogen (4.5 quality, Linde AS) at the anode and filtered compressed air at the cathode, with nitrogen for purging. Electrochemical characterisation was performed by use of a Biologic VMP3 with a 50 A FlexP0160 booster. The stack was operated at the conditions shown in Table 1 unless stated otherwise.

2.1. Conditioning protocol

Upon receipt of the stack, it was connected to the test bench, a leak test was conducted, and the stack was brought to nominal conditions,

Table 1

The operating conditions used in this work.

Condition		Unit	Value
Temperature	Coolant	°C	62
Relative Humidity	Anode	%	80
	Cathode	%	80
Backpressure	Anode	mbarg	200
	Cathode	mbarg	100
Stoichiometry	Anode	–	1.6
	Cathode	–	2
	Minimum stoichiometric current	A	70

then operated at 1 A cm^{−2} for 16 h. A performance harmonisation test was performed by measuring the voltage at a certain set of operating conditions to ensure the performance aligned with then manufacturer's expectations, then the initial characterisation was performed as outlined in 2.3.

2.2. Durability protocol

The testing protocol used in this work consisted of alternating 125 h periods of constant and dynamic current, with a full characterisation (as described in section 2.3) of stack performance at Beginning of Life (BOL) and after each 125 h period. The stack was operated for a total of 1500 h, after which the stack was shut down.

During constant current operation the stack was held at the relatively low current of 0.5 A cm^{−2} as this was deemed relevant for maritime operation due to high system efficiency resulting in minimised hydrogen costs [4]. A dynamic profile adapted from Bassam et al. [22] was used for the dynamic segments. This profile is based on the power cycle of the propulsion system of a 25.5 m long passenger vessel, FCS Alsterwasser, operating in Hamburg, Germany. Both profiles are shown in Fig. 1.

For the dynamic profile, the 360 s profile is repeated 1250 times during the cycling period of 125 h. The nominal current of the stack was chosen as the maximum current for the dynamic profile, while the minimum current was determined as the current at which the mean cell voltage dropped below 0.8 V in the BOL polarisation curve. The average current of the dynamic profile was 0.5 A cm^{−2}, meaning the total charge passed are the same for the constant and dynamic profiles. This allows direct comparison between degradation rates after each segment.

2.3. Characterisation

The stack was characterised every 125 h using several methods in the following order:

- Cell voltage decay (CVD) measurements were used to estimate hydrogen crossover in a non-quantitative manner. The measurement was performed at nominal conditions with hydrogen at the anode and no gas flow at the cathode. After the cathode flow was stopped, the time taken for the cell voltage to drop from 0.8 to 0.1 V was

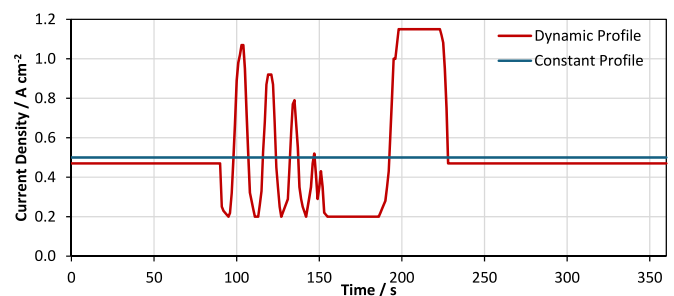


Fig. 1. The dynamic (red) and constant (blue) current profiles used in this work.

recorded and is determined to be proportional to the rate of hydrogen crossover.

- Polarisation curve (PC) measurements were used to define the overall stack performance and the magnitude of irreversible degradation. The PC was recorded at nominal conditions with a current range from 0 to 2.6 A cm^{-2} . The current steps were 0.05 A cm^{-2} with a 120 s hold time up to 0.2 A cm^{-2} , then 0.2 A cm^{-2} increments and a 300 s measurement time up to 2.6 A cm^{-2} .
- Electrochemical impedance spectroscopy (EIS) measurements were used to estimate the contributions to overall stack performance from the ohmic, activation and mass transport losses. The EIS was recorded at nominal conditions, at 0.2, 1.0 and 2.0 A cm^{-2} , with an AC perturbation of 5 % of the current setpoint. The measurement was performed from 10 kHz to 0.1 Hz and fitted to an R (R CPE) (R CPE) equivalent circuit using Relaxis software after confirmation that data meets the Kramers-Kronig transformation threshold of under 1 %.
- Charging current (CC) measurements as described by Brightman et al. [23] were used to determine the electrochemical active surface area (ECSA) of the cathode catalyst layer. Correction of the contributions of the double layer capacitance and hydrogen crossover were performed as outlined in Hu et al. [24] The CC measurement was performed at nominal temperature with 50 % H₂, 50 % N₂ content on the anode, and 100 % N₂ at the cathode with a relative humidity of 100 %, and flow rates of 4 and 8 NLPM respectively.

The BOL and End of Test (EOT) characterisation steps were repeated sequentially three times to estimate a standard deviation in the measurements.

2.4. Data analysis

Measured polarisation curves were fitted to a semi-empirical equation for the polarisation curve, both with and without mass transfer limitations, and on both individual cell level, and on stack level. The fitting parameters were the Tafel slope, b , exchange current density, i_0 , area specific resistance, R , transport coefficient, B , and lastly the limiting current density, i_L . The reversible potential was assumed to be known. However, due to very high uncertainties from the fitting procedure, most likely due to too many fitting parameters, the cell and stack area specific resistance was extracted from the EIS measurements as the real resistance at a frequency of 1000 Hz. Equation (1) shows the semi-empirical polarisation curve with mass transfer limitations for a cell, and its shortened version, for the entire stack, is shown in Equation (2).

$$E_{\text{Cell}}(i) = E^0 - \frac{RT}{nF} \ln \left\{ \frac{a_{\text{H}_2\text{O}}}{a_{\text{H}_2} \sqrt{a_{\text{O}_2}}} \right\} - b_{\text{Cell}} \log \left\{ \frac{i}{i_{0,\text{Cell}}} \right\} - iR_{\text{Cell}} + B_{\text{Cell}} \log \left\{ 1 - \frac{i}{i_{L,\text{Cell}}} \right\} \quad (1)$$

$$E_{\text{Stack}}(i) = E_{\text{Stack}}^{\text{Rev}} - b_{\text{Stack}} \log \left\{ \frac{i}{i_0} \right\} - iR_{\text{Stack}} + B_{\text{Stack}} \log \left\{ 1 - \frac{i}{i_L} \right\} \quad (2)$$

Each polarisation curve was fitted using a non-linear least squares approach to potential and current density data using a ‘Trust Region Reflective’ algorithm. This was implemented by using the `scipy.optimize.curve_fit` library in Python with the ‘trf’ method. The fit validation and fitting parameters are outlined in the [Supplementary Information Fig. S1](#) and [Table S1](#), respectively.

3. Results and discussion

3.1. Separation of the irreversible and reversible performance loss

The total performance loss over each measurement segment was determined by calculating the slope of the linear region of the cell voltage transient over time, as seen for the constant segments in [Fig. 2a](#), and for all dynamic segments in [Fig. 2b](#). For the constant current segments, the cell voltage was recorded every 60 s, and for the dynamic segments, the cell voltage was measured at the end of each cycle, resulting in one measurement every 360 s.

For both the dynamic and constant segments shown in [Fig. 2](#), it can be seen that there is a sharp decrease in cell voltage during the first hours of the segment before the decay enters a linear region. This can be attributed to the formation of PtOx at the catalyst, which reduces the active area available for ORR [19]. [Table 2](#) shows the total degradation rate calculated for each segment from the cell voltage transients, as well as the reversible and irreversible contributions.

The irreversible component of the degradation is obtained from the polarisation curves ([Fig. 3](#)), specifically, the change in mean cell voltage at 0.5 A cm^{-2} is divided by the segment time. It is assumed that the polarisation curve does not include any reversible degradation as two mitigation strategies were employed before the measurement. Firstly, any platinum oxidation is mitigated by a potential cycle that takes place before the CVD measurement. It has been shown that a single potential cycle to below 0.5 V is sufficient to reactivate the platinum surface of the catalyst and recover performance losses [25]. Secondly, buildup of water on the catalyst layer is mitigated during the CVD measurement, where there is no current applied i.e. no water is generated at the catalyst layer, and relatively high flow rates on the anode and cathode

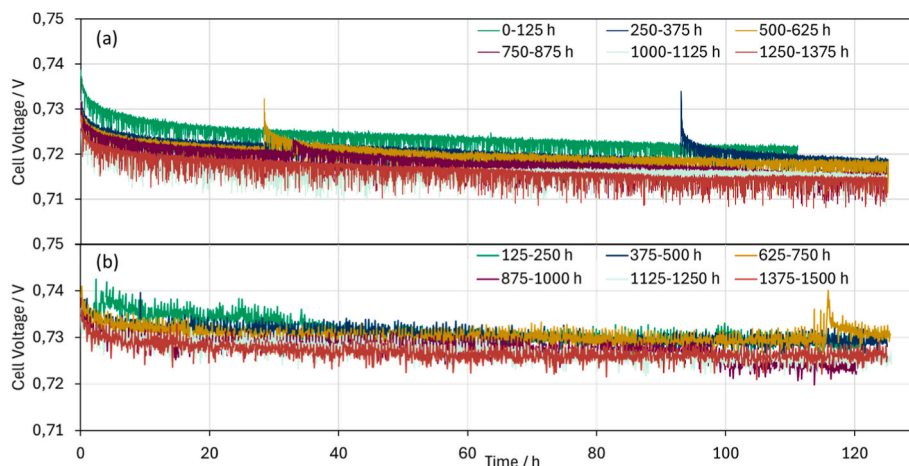


Fig. 2. The cell voltage transient for each segment of (a) constant current operation and (b) dynamic operation at the average current density of 0.5 A cm^{-2} .

Table 2

The degradation rates for each measurement segment. Total degradation rate determined from the cell voltage transient during each segment, irreversible degradation rate determined from the polarisation curve at 0.5 A cm^{-2} , and the reversible degradation rate determined from subtracting the irreversible from the total.

Elapsed Time/h	Cycle Type	Total Degradation Rate/ $\mu\text{V h}^{-1}$	Irreversible Degradation Rate/ $\mu\text{V h}^{-1}$	Reversible Degradation Rate/ $\mu\text{V h}^{-1}$
0000	–	–	–	–
0125	Constant	58	18	40
0250	Dynamic	61	40	21
0375	Constant	69	2	67
0500	Dynamic	30	0	30
0625	Constant	60	–2	62
0750	Dynamic	20	–2	22
0875	Constant	61	24	37
1000	Dynamic	25	18	07
1125	Constant	28	–16	44
1250	Dynamic	37	6	31
1375	Constant	49	–2	51
1500	Dynamic	23	2	21

expel any residual water accumulated during operation. Mitigation of the reversible degradation through application of a start-up/shutdown cycle was avoided as this has been shown to cause significant irreversible degradation [26,27].

Fig. 3 (a) displays the polarisation curves for each characterisation, and Table 2 shows the irreversible degradation rate at 0.5 A cm^{-2} for each segment. It should be noted that the irreversible degradation rate varies at different current setpoints, with an increased degradation rate seen at higher current setpoints, as in agreement with literature [28]. At higher currents, losses of all types (activation, ohmic, mass transport) are combined, in our case, at 1 A cm^{-2} the total degradation during 1500 h of testing is around 12 mV, which increases to 29 mV at 2.6 A cm^{-2} . The rates in Table 2 are reported at 0.5 A cm^{-2} , the average current applied during testing, to allow separation of the reversible and irreversible contributions.

It can be observed that the amount of irreversible degradation is highest during the first 250 h of operation, before levelling off to around zero micro volts per hour between 250 and 750 h. The initially high irreversible degradation rate is commonly seen in literature and can be attributed to initial rapid catalyst particle size growth (Ostwald ripening) [29–31].

The degradation rate is once again elevated between 750 and 1000 h of operation. This could possibly be due to variations in the quality of the hydrogen supply. For these measurements, the hydrogen quality used is 4.5 (i.e. 99.995 % purity), and approximately one pack per week is consumed. This 4.5 quality H_2 can have a combined CO and CO_2 content of up to 2 ppm, quantities that have been shown to reduce performance significantly [32,33]. This will cause an elevated rate of reversible degradation that gradually recovers when the source of CO is stopped (i.e. a new pack with less contamination is used) and oxygen crossing over from the cathode oxidises any adsorbed CO to CO_2 , removing it from the catalyst layer. There is a small amount of Ruthenium in the anode catalyst, which should mitigate some of the impact of CO on performance loss, and thus facilitate an easier oxidation of CO to CO_2 . However, this will not completely stop the poisoning effect. Part of the incurred degradation is recovered during the segment from 1000 to 1125 h, as would be expected for CO poisoning, however other possible explanations for this behaviour cannot be excluded. After 1125 h the irreversible degradation rate once again becomes negligible.

Finally, the irreversible degradation is subtracted from the total amount of degradation to obtain the amount of reversible degradation, with values shown in Table 2. The magnitude of reversible degradation varies between 7 and $67 \mu\text{V h}^{-1}$ and does not show any trends related to the elapsed time. Differences in degradation rate between the constant and dynamic segments are explored in Section 3.4.

3.2. Determination of the source of irreversible performance losses

Although the total irreversible reduction in performance during the 1500 h testing time was relatively small at only 11 mV, it is important to understand the reasons behind this, to identify whether improvements in material selection or changes in operation strategy can be made. Thus, the electrochemical techniques of CVD, EIS, and CC, were utilised, as well as PC fitting.

3.2.1. Overall losses

The polarisation curve can be used to quantify how the different types of potential losses contribute to the overall potential of the stack (and cell) for different current densities. Fig. 4c shows the magnitude of the activation loss, ohmic loss, and transport loss against time at a current density of 2.6 A cm^{-2} . The fitted ohmic resistance remains mostly constant over time, implying little change in the ionic conductivity of the membrane, in agreement with HFR measurements (Section

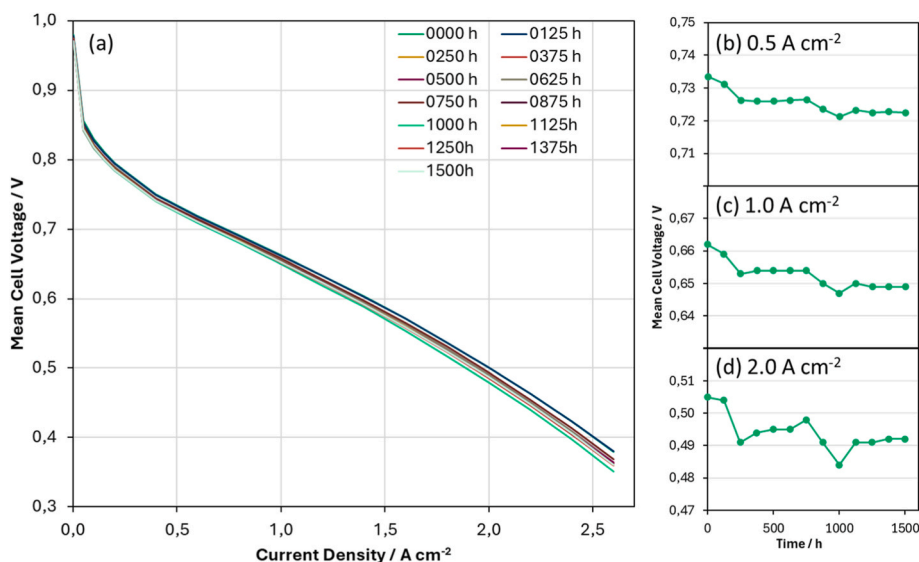


Fig. 3. (a) Polarisation curves showing the mean cell voltage at various current densities at each characterisation point. The insets show the progression of the mean cell voltage over time at (b) 0.5 A cm^{-2} , (c) 1.0 A cm^{-2} , and (d) 2.0 A cm^{-2} .

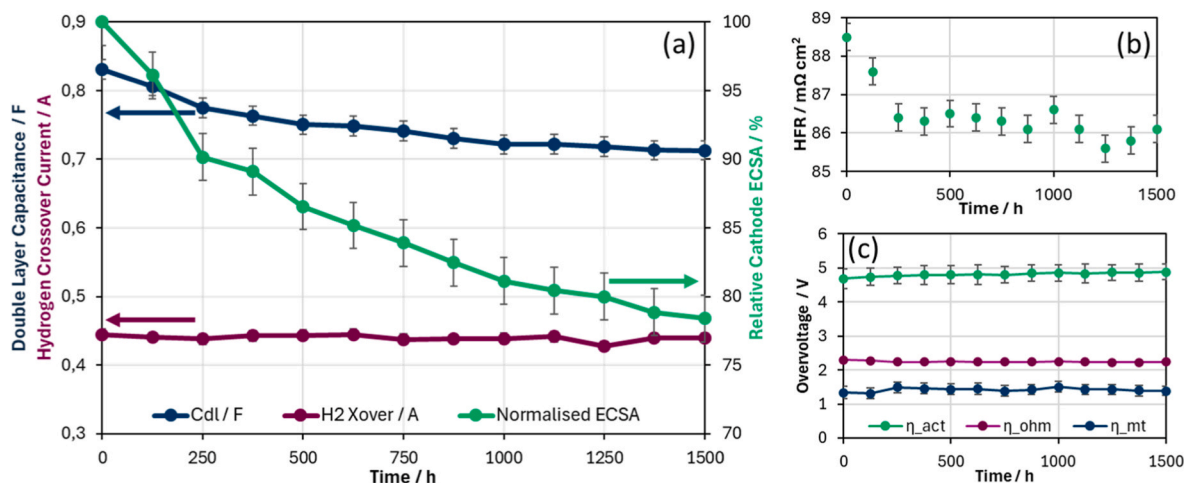


Fig. 4. (a) The relative change in cathode ECSA, hydrogen crossover, and double layer capacitance over time for the stack. (b) The change in HFR over time for the stack. (c) The voltage losses over time due to activation (η^{act}), ohmic resistance (η^{ohm}), and transport (η^{mt}) overpotentials at 2.6 A cm^{-2} .

3.2.4.

At these currents, which are over two times higher than nominal, it could be expected that mass transport limitations dominate, but it was found that mass transport losses do not contribute significantly to overall performance loss, with the limiting current density found by the fitting procedure to be between 2.77 A cm^{-2} and 3.28 A cm^{-2} . Instead, Fig. 4c shows that the activation overpotential was the biggest source for potential loss, and this increased linearly over time, due to a 20 % reduction in ECSA (Section 3.2.2).

3.2.2. Electrochemical catalytic surface area

Fig. 4 (a) shows the relative cathode ECSA change during testing. It is expected that the majority of the degradation occurs on the cathode due to the higher potentials and oxidative environment. The ECSA of the anode catalyst was thus not recorded.

During the first 250 h, Ostwald ripening is the dominant process for ECSA loss [29], which in this case causes a 10 % drop in ECSA. After this, there is a linear decrease in ECSA of around 1 % per 125 h segment, with a final active catalyst area of around 79 % of the original. A similar trend in the development of the ECSA of a PEMFC stack during 2000 h of testing has been reported in literature by Robin et al., [34] as well as on the single cell by Colombo et al., [28] amongst others. This decrease in ECSA can be due to a number of phenomena, including detachment or agglomeration of catalyst particles, or corrosion of the carbon support. For the work presented here, it is assumed that the catalyst support corrosion is negligible as the potential during this work is at all times is below the potential at which carbon oxidation is onset, 0.85 V, and significantly lower than the potential at which it is kinetically favoured, >1 V. However, other types of degradation such as Pt migration and agglomeration will be constantly occurring. The double layer capacitance (C_{DL}) also decreases in a similar trend to the ECSA, as expected due to the CDL being directly proportional to the electrode surface area [35].

Despite a 20 % drop in ECSA during the testing period, there is only a relatively small drop in cell voltage. In this case, the commercial stack was designed for sustained performance over a long lifetime, meaning that the cathode catalyst loading is high, and therefore even with a 20 % reduction, the loading is still more than enough needed for high efficiency oxygen reduction. Other works have seen a similar phenomenon, where a significant decrease in ECSA is needed before voltage drops are observed, for example in Ref. [28] a 40 % ECSA loss over 1000 h of dynamic operation leads to only 80 mV performance drop.

3.2.3. Hydrogen crossover

The development of the relative hydrogen crossover over time is shown in Fig. 4 (a). The rate of hydrogen crossover can be an indicator of

membrane health, as a membrane that is thinning or has pinholes will show a significantly higher rate of crossover, indicating degradation. Fig. 4 shows no significant increase in hydrogen crossover throughout the test. The hydrogen crossover current is around 0.45 A, which is well under the threshold value of 2 A that indicates a damaged membrane that must be replaced.

3.2.4. Electrochemical impedance spectroscopy analysis

EIS analysis of the stack was performed after each 125 h of operation, and the obtained spectra can be seen in the [Supplementary Information Fig. S2](#), with the development of the high-frequency intercept throughout the testing duration shown in Fig. 4 (b). After an initial decrease in HFR during the first 250 h, it remains relatively constant for the rest of the test, at around 86 mΩ cm^2 . Together with the lack of change in hydrogen crossover, this shows that there is no membrane degradation observed during the 1500 h of maritime relevant testing.

3.3. Separation of the individual cell performance

The standard deviation of the cells in the stack was used as a metric to determine how the relative degradation changed over time. Fig. 5 shows the difference of each cell from the mean at 0.5 A cm^{-2} at four time points during the testing. Similar data at additional current densities can be found in the [Supplementary Information Fig. S3](#).

Notably, the first and last cells have a lower than average performance throughout the entire testing duration. This is due to a lower temperature at these edge cells compared to the bulk of the stack, leading to lower gas diffusivity and thus membrane conductivity,

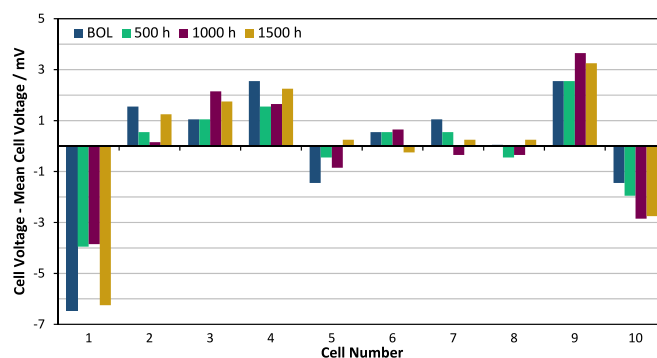


Fig. 5. The relative cell voltage at 0.5 A cm^{-2} compared to mean for each individual cell at four time points during testing.

resulting in a lower thermodynamic efficiency.

Otherwise, none of the cells in the stack have a sharp increase or decrease in relative performance, which indicates consistent degradation over the testing time. The maximum difference between the best and worst performing cells is less than 10 mV during the entire testing time. Similarly, looking at the relative ECSA of the individual cells (Supplementary Information Fig. S4) shows that the final relative ECSA varies very little, between 79.3 % (cell 1) and 80.9 % (cell 2) of the BOL value. There seems to be a relatively even ECSA decline over the whole stack, with no notable increase in the spread of cells at the end of test.

It is to be expected that a significantly longer test duration would result in a larger variation in cell performance developing over time [36, 37]. This would be a result of inconsistent degradation which could be attributed to small variations in the manufacturing of the components, as well as differences in the conditions experienced by each cell with respect to heat and water production.

3.4. The influence of dynamic operation

This work aims to understand the impact of dynamic operation at maritime relevant current densities on fuel cell degradation, therefore two load profiles of the same average current density were applied to the stack, and the reversible and irreversible degradation quantified.

Fig. 6 shows the average reversible and irreversible losses for the constant and dynamic segments of the fuel cell stack operation, ignoring the elevated degradation rates during the first 250 h of operation. Interestingly, the overall degradation rate for the constant segments is higher than for the dynamic segments. This is due to the significant contribution of reversible losses to the overall voltage decay rate, with constant segments having around $52 \mu\text{V h}^{-1}$ of reversible losses compared to $22 \mu\text{V h}^{-1}$ for the dynamic segments. The values for total degradation are in line with those observed by Chu et al., [17] who reported a total degradation rate of $42 \mu\text{V h}^{-1}$ during steady state operation at slightly higher operational currents of 1.22 A cm^{-2} (mean cell voltages from 0.65 to 0.60 V).

One explanation for this elevated reversible voltage decay rate during the constant segments can be attributed to a lack of changes in current leading to constant catalyst layer potential and gas flow rates, thus neither of the main strategies for mitigating reversible degradation are in place (low voltage setpoints to reduce any Pt-oxides, and variable gas flow rates that assist in removal of water buildup from the catalyst layer). Kim et al. [38] also found that at low load conditions less than 0.5 A cm^{-2} , static operation led to higher degradation rates than dynamic operation, although they did not quantify whether this

degradation was reversible or not. This interesting finding indicates that even hybridised fuel cell systems that follow a load levelling energy management strategy will be prone to high levels of degradation when the reversible component is not properly mitigated.

However, even though the reversible losses are higher for the constant segments, the irreversible losses are higher for the dynamic segments than for the constant segments, with values of 4.8 and $1.2 \mu\text{V h}^{-1}$ respectively. To understand this, Table 3 shows rates of change of key performance metrics separated by segment type.

Firstly, the ECSA decrease per segment was slightly larger for the dynamic segments compared to the constant segments. This can be due to variable heat and water production which can exacerbate ECSA loss [39]. In contrast, the hydrogen crossover increased for the constant segments but decreased for the dynamic segments. This may be because H_2 crossover is directly influenced by membrane humidity, where the constant segments allowed for a greater membrane humidity by avoiding periods of high gas stoichiometry and low currents, which dry the membrane. In addition, the HFR showed a small increase for the dynamic segments compared to a small decrease for the constant segments, however this increase is within the measurement uncertainty therefore may not be a significant contributor to the overall voltage loss. Thus, the main reason for increased irreversible degradation during dynamic operation is an increased rate of ECSA loss.

3.5. Discussion

The irreversible degradation rates reported in this study are overall relatively low, particularly for the constant cycles. The total irreversible degradation in cell voltage was only 11 mV over the course of the testing, despite an ECSA reduction of around 21 %. This illustrates that commercial PEMFC stacks often have an excess catalyst loading that can allow such a loss of active area without a large influence on total stack performance. This emphasises that for applications with long required lifetimes such as maritime, the priority for manufacturers is to achieve lifetime targets rather than to reduce initial CAPEX. In this case, if the lifetime of the stack is defined as the time taken for a 10 % reduction in performance, the predicted lifetime for constant operation with mitigation of reversible losses is up to 60 000 h. This is approaching the requirements for maritime vessels, 80 000 h, and is significantly longer than automotive targets of 30 000 h. However, if mitigation strategies are not taken into account, this lifetime drops to only 1350 h, visualising the importance of understanding the degradation processes occurring within the fuel cell.

Additionally, these results show that dynamic operation is worse for fuel cell lifetime than constant operation, meaning that hybridisation of the fuel cell system is recommended, even for maritime applications that may have a lower average current density and less dynamic load than automotive applications. However, even for hybrid systems where the fuel cell load profile is relatively constant, mitigation of the reversible degradation is of great importance, and must be performed frequently if long-term operation is needed. Mitigation strategies for recovering reversible performance losses can include potential cycling to clean the catalyst layer, reducing temperature to remove water buildup, or shutdown/start-up cycling, amongst others [16,21].

Table 3

An overview of the contributions to the irreversible performance losses for each segment type. The values are all represented on a per-segment basis, with the segment length of 125 h. The data excludes the first 250 h of operation.

Segment Type	Irreversible losses at $0.5 \text{ A cm}^{-2}/\mu\text{V h}^{-1}$	Cathode ECSA decrease/% per segment	Increase in H_2 crossover/ mA per segment	Change in HFR/ $\text{m}\Omega \text{ cm}^2 \text{ per segment}$
Constant	1.2	1.3	0.8	-0.11
Dynamic	4.8	1.6	-1.6	+0.16

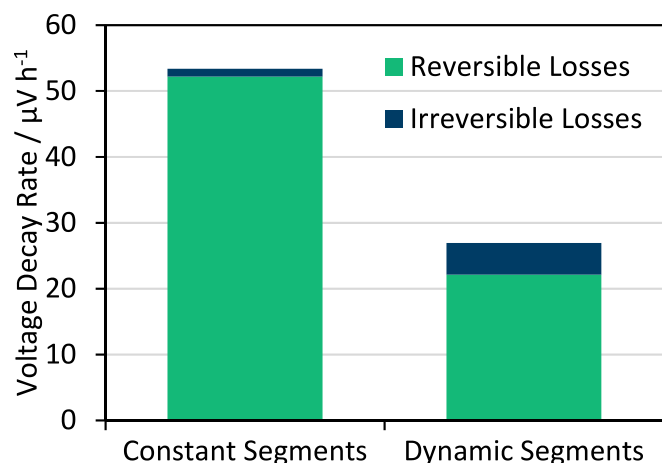


Fig. 6. The total voltage decay rate observed for each type of segment, excluding the first 250 h of operation, split into the contributions from reversible (green) and irreversible (blue) losses.

4. Conclusion

A commercial fuel cell short stack was tested for 1500 h by applying alternating periods of constant load and a dynamic maritime load profile. During this time, a mean cell voltage reduction of 11 mV was observed, primarily due to a reduction in ECSA of around 21 %. The majority of this degradation (7 mV, 10 % ECSA loss) occurred during the first 250 h of operation and can be attributed to Ostwald ripening of the catalyst particles. During the testing, no membrane degradation was observed, with a constant hydrogen crossover rate and HFR.

Comparing the constant and dynamic load profiles, the dynamic profile led to significantly reduced reversible degradation, but an increased amount of irreversible degradation compared to the constant profile. This was found to be due to an increased ECSA loss during the dynamic segments. For both cycle types, reversible degradation dominated, meaning that it should be continuously monitored and mitigated at regular intervals, regardless of cycle type.

This work indicates that current commercial fuel cell stacks can be suitable for use in a maritime environment, with indicative lifetimes approaching the 80 000 h targets, provided that hybridisation is applied to allow constant operation, and that reversible degradation is periodically mitigated.

CRediT authorship contribution statement

Katie McCay: Writing – original draft, Visualization, Methodology, Investigation, Formal analysis, Conceptualization. **Pål Emil England Karstensen:** Writing – original draft, Visualization, Formal analysis. **Michael Gerhardt:** Writing – review & editing, Validation, Formal analysis. **Jorg Coolegem:** Writing – review & editing, Resources, Methodology. **Danilo Damiano:** Writing – review & editing, Resources, Methodology. **Sigrid Lædre:** Writing – review & editing, Project administration, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.jpowsour.2025.239011>.

Data availability

No data was used for the research described in the article.

References

- [1] International Energy Agency, *World Energy Outlook 2024*, 2024.
- [2] International Energy Agency: International Shipping," <https://www.iea.org/energy-system/transport/international-shipping>.
- [3] IMO, 2023 IMO Strategy on Reduction of GHG Emission from Ships, 2023.

- [4] D.A. Cullen, et al., New Roads and Challenges for Fuel Cells in heavy-duty Transportation, 2021, <https://doi.org/10.1038/s41560-021-00775-z>.
- [5] Z. Wang, M. Li, F. Zhao, Y. Ji, F. Han, Status and prospects in technical standards of hydrogen-powered ships for advancing maritime zero-carbon transformation, *Int. J. Hydrogen Energy* 62 (Apr. 2024) 925–946, <https://doi.org/10.1016/j.IJHYDENE.2024.03.083>.
- [6] M. de las, N. Camacho, D. Jurburg, M. Tanco, Hydrogen fuel cell heavy-duty trucks: review of main research topics, *Int. J. Hydrogen Energy* 47 (68) (Aug. 2022) 29505–29525, <https://doi.org/10.1016/j.IJHYDENE.2022.06.271>.
- [7] R. Knibbe, et al., Optimal battery and hydrogen fuel cell sizing in heavy-haul locomotives, *J. Energy Storage* 71 (Nov. 2023) 108090, <https://doi.org/10.1016/J.EST.2023.108090>.
- [8] Clean Hydrogen Joint Undertaking, *Strategic Research and Innovation Agenda 2021 - 2027*, 2022.
- [9] H. Xing, C. Stuart, S. Spence, H. Chen, Fuel cell power systems for maritime applications: progress and perspectives, *Sustainability* 13 (3) (Jan. 2021) 1213, <https://doi.org/10.3390/su13031213>.
- [10] H. Lohse-Busch, et al., Automotive fuel cell stack and system efficiency and fuel consumption based on vehicle testing on a chassis dynamometer at minus 18 °C to positive 35 °C temperatures, *Int. J. Hydrogen Energy* 45 (1) (2020), <https://doi.org/10.1016/j.ijhydene.2019.10.150>.
- [11] L. Balestra, I. Schjøberg, Energy management strategies for a zero-emission hybrid domestic ferry, *Int. J. Hydrogen Energy* 46 (77) (Nov. 2021) 38490–38503, <https://doi.org/10.1016/j.IJHYDENE.2021.09.091>.
- [12] G. Smith, K. McCay, Fuel cell short stack testing, in: *Fuel Cells for Transportation: Fundamental Principles and Applications*, 2023, <https://doi.org/10.1016/B978-0-323-99485-9.00010-1>.
- [13] P. Gazdzick, J. Mitzel, D. Garcia Sanchez, M. Schulze, K.A. Friedrich, Evaluation of reversible and irreversible degradation rates of polymer electrolyte membrane fuel cells tested in automotive conditions, *J. Power Sources* 327 (2016), <https://doi.org/10.1016/j.jpowsour.2016.07.049>.
- [14] G. De Marco, T. Malkow, G. Tsotridis, A. Pilenga, EU Harmonised Test Protocols for PEMFC MEA Testing in Single Cell Configuration for Automotive Applications, 2015, <https://doi.org/10.2790/54653>.
- [15] A. Broer, H. Polinder, L. van Biert, Polymer electrolyte membrane fuel cell degradation in ships — review of degradation mechanisms and research gaps, *J. Power Sources* 640 (Jun. 2025) 236678, <https://doi.org/10.1016/J.JPOWSOUR.2025.236678>.
- [16] J. Mitzel, Q. Zhang, P. Gazdzick, K.A. Friedrich, Review on mechanisms and recovery procedures for reversible performance losses in polymer electrolyte membrane fuel cells, *J. Power Sources* 488 (Mar. 2021) 229375, <https://doi.org/10.1016/J.JPOWSOUR.2020.229375>.
- [17] T. Chu, et al., Investigation of the reversible performance degradation mechanism of the PEMFC stack during long-term durability test, *Energy* 258 (2022), <https://doi.org/10.1016/j.energy.2022.124747>.
- [18] J. Zhao, X. Li, A Review of Polymer Electrolyte Membrane Fuel Cell Durability for Vehicular Applications: Degradation Modes and Experimental Techniques, 2019, <https://doi.org/10.1016/j.enconman.2019.112022>.
- [19] M. Zago, et al., Experimental analysis of recoverable performance loss induced by platinum oxide formation at the polymer electrolyte membrane fuel cell cathode, *J. Power Sources* 455 (Apr. 2020) 227990, <https://doi.org/10.1016/J.JPOWSOUR.2020.227990>.
- [20] T. Ma, et al., Recovery characteristics of reversible degradation for proton exchange membrane fuel cell stack under accelerated stress test, *Chem. Eng. J.* 493 (Aug. 2024) 152549, <https://doi.org/10.1016/J.CEJ.2024.152549>.
- [21] R. Li, et al., Investigation on performance recovery and consistency of proton exchange membrane fuel cell stack under automotive accelerated stress test, *Appl. Energy* 383 (Apr. 2025) 125388, <https://doi.org/10.1016/J.APENERGY.2025.125388>.
- [22] A.M. Bassam, A.B. Phillips, S.R. Turnock, P.A. Wilson, Development of a multi-scheme energy management strategy for a hybrid fuel cell driven passenger ship, *Int. J. Hydrogen Energy* 42 (1) (Jan. 2017) 623–635, <https://doi.org/10.1016/J.IJHYDENE.2016.08.209>.
- [23] E. Brightman, G. Hinds, R. O'Malley, In situ measurement of active catalyst surface area in fuel cell stacks, *J. Power Sources* 242 (Nov. 2013) 244–254, <https://doi.org/10.1016/J.JPOWSOUR.2013.05.046>.
- [24] Z. Hu, et al., Comprehensive analysis of galvanostatic charge method for fuel cell degradation diagnosis, *Appl. Energy* 212 (Feb. 2018) 1321–1332, <https://doi.org/10.1016/J.APENERGY.2018.01.005>.
- [25] F.A. Uribe, T.A. Zawodzinski, A study of polymer electrolyte fuel cell performance at high voltages. Dependence on cathode catalyst layer composition and on voltage conditioning, *Electrochim. Acta* 47 (22–23) (Aug. 2002) 3799–3806, [https://doi.org/10.1016/S0013-4686\(02\)00350-X](https://doi.org/10.1016/S0013-4686(02)00350-X).
- [26] T. Zhang, P. Wang, H. Chen, P. Pei, A review of automotive proton exchange membrane fuel cell degradation under start-stop operating condition, *Appl. Energy* 223 (Aug. 2018) 249–262, <https://doi.org/10.1016/J.APENERGY.2018.04.049>.
- [27] T. Fletcher, RobThring, M. Watkinson, An energy management strategy to concurrently optimise fuel consumption & PEM fuel cell lifetime in a hybrid vehicle, *Int. J. Hydrogen Energy* 41 (2016) 21503–21515.
- [28] E. Colombo, A. Baricci, A. Bisello, L. Guetaz, A. Casalegno, PEMFC performance decay during real-world automotive operation: evincing degradation mechanisms and heterogeneity of ageing, *J. Power Sources* 553 (Jan. 2023) 232246, <https://doi.org/10.1016/J.JPOWSOUR.2022.232246>.
- [29] R. Sharma, S. Gyergyek, Q. Li, S.M. Andersen, Evolution of the degradation mechanisms with the number of stress cycles during an accelerated stress test of

- carbon supported platinum nanoparticles, *J. Electroanal. Chem.* 838 (Apr. 2019) 82–88, <https://doi.org/10.1016/J.JELECHEM.2019.02.052>.
- [30] M. Cong, K. Wang, N. Yao, T. Ma, Study on the degradation of proton exchange membrane fuel cell under load cycling conditions, *Int. J. Hydrogen Energy* 47 (91) (Nov. 2022) 38736–38746, <https://doi.org/10.1016/J.IJHYDENE.2022.09.032>.
- [31] Y. Zhao, M. Luo, J. Yang, B. Chen, P.-C. Sui, W. Zhou, Experimental study of PEMFC stack performance degradation considering the shutdown and rest processes, *Int. J. Energy Res.* 2025 (1) (Jan. 2025), <https://doi.org/10.1155/er/3246950>.
- [32] B. Shabani, M. Hafttananian, Sh Khamani, A. Ramiar, A.A. Ranjbar, Poisoning of proton exchange membrane fuel cells by contaminants and impurities: review of mechanisms, effects, and mitigation strategies, *J. Power Sources* 427 (2019) 21–48.
- [33] X. Cheng, et al., A review of PEM hydrogen fuel cell contamination: impacts, mechanisms, and mitigation, *J. Power Sources* 165 (2) (Mar. 2007) 739–756, <https://doi.org/10.1016/J.JPOWSOUR.2006.12.012>.
- [34] C. Robin, M. Gérard, M. Quinaud, J. d'Arbigny, Y. Bultel, Proton exchange membrane fuel cell model for aging predictions: simulated equivalent active surface area loss and comparisons with durability tests, *J. Power Sources* 326 (2016), <https://doi.org/10.1016/j.jpowsour.2016.07.018>.
- [35] J. Larminie, A. Dicks, *Fuel Cell Systems Explained*, second ed., 2013, <https://doi.org/10.1002/9781118878330>.
- [36] X. Zhang, Y. Jiang, L. Huang, W. Chen, D. Brett, Inconsistent responses of cells on operating conditions in a 5 kW proton exchange membrane fuel cell stack, *Electrochim. Acta* 391 (2021), <https://doi.org/10.1016/j.electacta.2021.138925>.
- [37] S. Murata, M. Imanishi, S. Hasegawa, R. Namba, Post-mortem analysis of a long-term tested proton exchange membrane fuel cell stack under low cathode humidification conditions, *J. Power Sources* 253 (May 2014) 90–97, <https://doi.org/10.1016/J.JPOWSOUR.2013.11.056>.
- [38] J.H. Kim, et al., Analysis of the impact of dynamic operation of PEMFCs for ship applications, *J. Ocean Eng. Technol.* 38 (5) (2024) 199–209.
- [39] R.L. Borup, et al., Recent developments in catalyst-related PEM fuel cell durability, *Curr. Opin. Electrochem.* 21 (Jun. 2020) 192–200, <https://doi.org/10.1016/J.COEELEC.2020.02.007>.